

AI Medical Report Analyzer (React + FastAPI) 4-Week Development Roadmap

This roadmap is designed for a first-year bioinformatics student who already knows Python basics and wants to build a strong internship-level healthcare AI project using React and FastAPI.

Week 1 — Backend Foundations + PDF Processing

- Set up project structure and virtual environment
- Install FastAPI, Uvicorn, pdfplumber, pandas
- Build FastAPI backend and health-check API
- Implement PDF upload endpoint
- Extract text from PDFs using pdfplumber
- Learn OCR basics using pytesseract and Pillow
- Refactor backend into modular architecture

Week 2 — Medical Analysis Engine

- Learn regex properly for text extraction
- Build medical parameter extractor
- Create normal range comparison system
- Implement HIGH / LOW / NORMAL classification
- Generate medical summaries
- Improve extraction accuracy for multiple report formats
- Integrate analysis into backend API

Week 3 — React Frontend + UI

- Create React app using Vite
- Install Axios, React Router, Tailwind CSS
- Configure Tailwind styling
- Build file upload component

- Connect React frontend to FastAPI backend
- Display extracted results and summaries
- Improve UI using cards, loaders, and colors

Week 4 — AI Features + Deployment

- Integrate OpenAI API for AI-generated summaries
- Improve prompts for patient-friendly explanations
- Add error handling
- Initialize GitHub repository
- Deploy backend on Render
- Deploy frontend on Vercel
- Add README, screenshots, and documentation

Final MVP Architecture

React Frontend → Axios Requests → FastAPI Backend → PDF/OCR Processing → Medical Parameter Extraction → Analysis Engine → AI Summary Generation → Frontend Dashboard

Minimum Viable MVP Checklist

- ✓ FastAPI backend
- ✓ PDF upload endpoint
- ✓ PDF extraction
- ✓ Medical parameter extraction
- ✓ Medical analysis engine
- ✓ React frontend
- ✓ API integration
- ✓ AI summary generation
- ✓ GitHub repository
- ✓ Deployment

By the end of these 4 weeks, you will have a strong healthcare AI portfolio project that demonstrates full-stack development, AI integration, medical data analysis, and bioinformatics relevance.